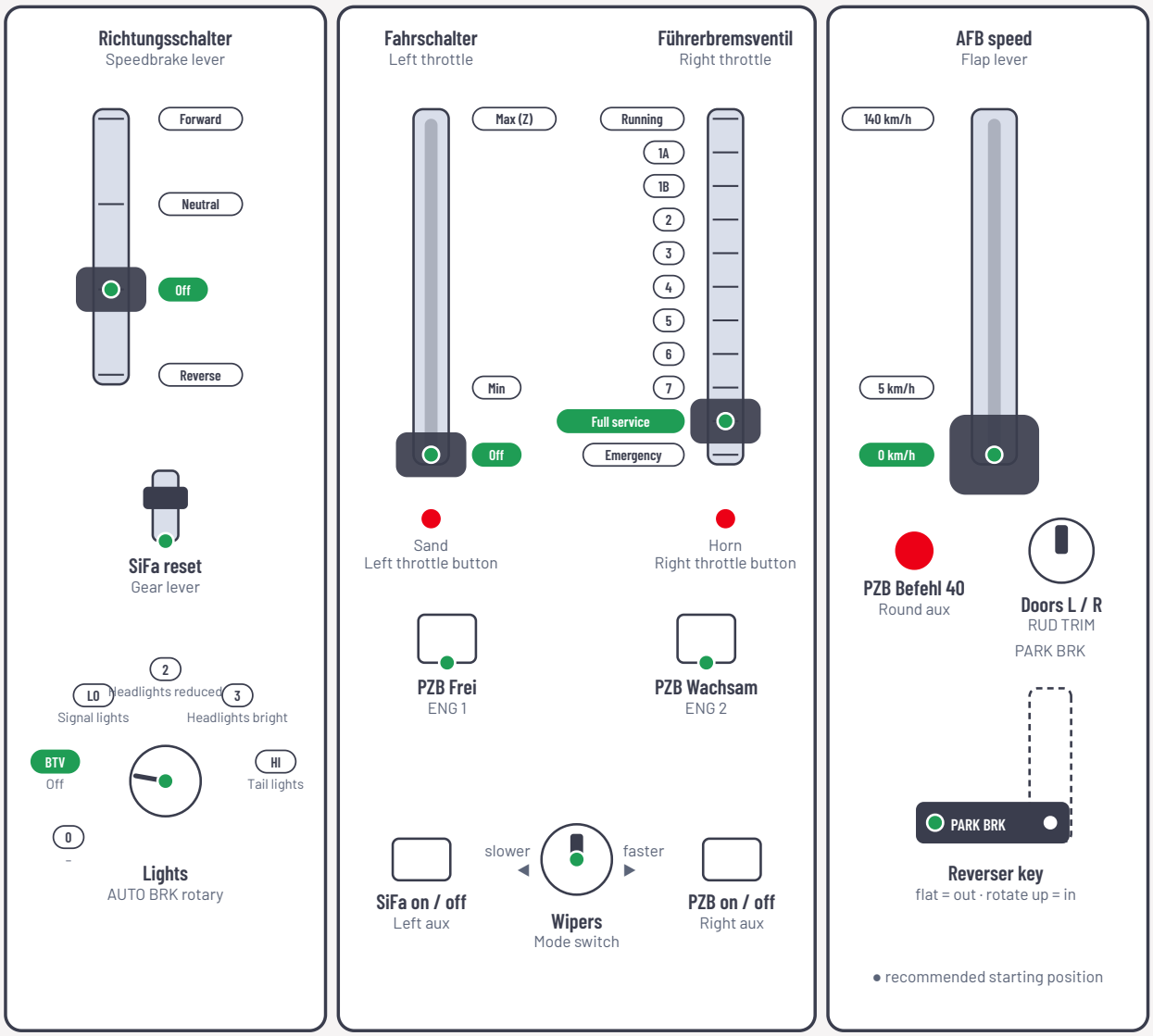
BR 145 Expert



Cab setup

- Battery on; Fahrschalter at Off; PARK BRK up: reverser key in (held while it is up)
- Richtungsschalter straight to V (Forward): this occupies the cab
- Machine room: breakers closed, SiFa and LZB/PZB cut-off valves open; Cab 2 checked (page 2)
- Raise the pantograph, close the Hauptschalter, air compressor on
- SiFa and PZB on (aux squares); brake preparation, PZB/LZB and SiFa tests, train data on the MTD (page 2)
- Lights to Signal lights or Headlights; AFB on in the cab for the speed setter

Cleared to depart

- Close doors, check the signal and PZB
- Parking brake off, Führerbremsventil to Running, release the direct brake
- Fahrschalter towards Z (it moves only at V), or set the AFB speed on the flap lever
- Acknowledge SiFa with the gear lever, PZB with the ENG switches
- Brake: Führerbremsventil towards SB in steps



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BR 145 Expert Aufrüsten: from cold to departure

A cold-start service finds the BR 145 dark; a timetable service loads it with no cab occupied ('Kein Führerstand besetzt'), SiFa isolated and its cut-off valves closed. The order: Cab 1, machine room, Cab 2, Cab 1 again.

Cab 1: battery and desk

1. Loco under the wire; at least one brake applied (red square with a black dot).
2. **CAB 1 BACK WALL** Battery (Batterie) on; every breaker on this wall closed, switches normal, auxiliary controls off.
3. Headlights as required. (AUTO BRK)
4. Fahrschalter at Off; reverser key in, and keep PARK BRK up: the key is held only while it is. (Left throttle) (PARK BRK)
5. Richtungsschalter to V (Forward); this occupies the cab. (Speedbrake lever)
6. **DESK** Signal and instrument lights on, AFB off; ease the direct brake to 0.5 bar C-pressure or less.
7. **BACK WALL** Parking brake on (light lit).

Machine room and Cab 2

8. **MACHINE ROOM** Close every LSS and MSS breaker in the HBG 1, FML 2 and HBG 2 cabinets.
9. **HBG 1** SiFa, LZB and PZB isolation switches to normal. (Left aux) (Right aux)
10. Open the SiFa and LZB/PZB cut-off valves; the pantograph cut-off valve open.
11. Leave the driver's brake cut-off valve closed, as the manual's start-up wants.
12. **CAB 2** Cab 2: breakers closed, switches normal, lights as required, direct brake at Hold, test the horn pedal.
13. **CAB 2** Cab 2: air compressor switch on, signal and instrument lights on; walk back to Cab 1.

Power up

14. **CAB 1 DESK** Raise the pantograph, close the Hauptschalter, switch the air compressor on.
15. **LEFT MTD** Train number: MTD [4] Zug-BeSy, [2] Zug-/Tf.Nr., [E] for the driver ID; set up GSM-R.

Brake preparation and tests

16. Führerbremsventil at Running; the brake pipe stays at 5 bar. (Right throttle)
17. Richtungsschalter to M (Neutral): still 5 bar; back to V. (Speedbrake lever)
18. Electric brake handle to Decouple (in the cab), brake pipe down 0.8 bar, release. (Right throttle)
19. **LEFT MTD** MTD W, Prüfen, soft key 5 'BrR 2 Ein'; after the HSM change repeat Decouple, 0.8 bar, release.
20. **DESK** Apply and release the direct brake, watching the C-pressure gauge.
21. **LEFT MTD** PZB/LZB test with a direction set and the train brake released: MTD [4] ZugBesy, [8] Prüfen.
22. **MFA** It shows the MFA lamps, two emergency applications that empty and refill the pipe, LZB 1200 / 1000 / 120 / 115.
23. **SIFA PEDAL** SiFa test: direct brake on, hold the pedal (Q) 35 s: light at 30 s, 'Sifa' three times, then emergency brake.
24. Direct brake on, release the parking brake, machine room and cab lights off, test the horn. (Right throttle button)
25. Rolling test: release all brakes, a little power, roll a few metres, power to 0, stop on the direct brake. (Left throttle)

Wagons: brake test (Bremsprobe)

26. Parking and hand brakes on, NBÜ/ep at 0, fill the brake pipe to 5 bar. (Right throttle)
27. **OUTSIDE** First wagon released: kick a brake shoe (it moves) or hammer a wheel (a bright, long ring).
28. Richtungsschalter to M cuts the brake valve off: at most 0.5 bar lost in 60 s (0.3 passenger). (Speedbrake lever)
29. Back to V, let the pipe reach 5.0 bar, then brake to 4.2 bar or lower. (Speedbrake lever) (Right throttle)
30. **OUTSIDE** Walk one side: brakes applied, couplers tight, hoses joined, cocks open, hand brakes off.
31. **LAST WAGON** Rear: angle cock closed, hose and coupler on their hooks, Zg 2 tail signals present.
32. Release to 5.0 bar; walk the other side: brakes released, brake mode and load as the Wagenliste says. (Right throttle)
33. **LAST WAGON** Continuity: open the rear angle cock 15 s; the last wagon brakes, then releases.

Train data (ZDE)

34. **BREMSZETTEL** BRA from the loco's brake mode: R = 9, P = 4, G = 2.
35. BRH: convert the Brh % with the BR 145 ZDE sheet (76–95 % gives 080).
36. ZL: length rounded up to tens (533 m = 54). VMZ: the formation limit, else the timetable speed.
37. MTD [4] ZugBesy, [1] ZDE: BRA, BRH, ZL, VMZ, [E] twice, then hold PZB Wachsam a second. (ENG 2)
38. Check: hold Wachsam and Frei together; [85] lights and the MFA shows the data. (ENG 1 + ENG 2)

Departure and Abrüsten

39. At V: parking brake off, Führerbremsventil to Running, release the direct brake. (Speedbrake lever) (Right throttle)
40. Fahrschalter through Min upwards; or AFB on, Fahrschalter back to 0, set the speed. (Left throttle) (Flap lever)
41. On the move: SiFa reset, PZB Wachsam, Frei, Befehl 40. (Gear lever) (ENG 2) (ENG 1) (Round aux)
42. Abrüsten: radio off or mode C, direct brake to 0.5 bar, parking brake on, sand both ways. (Left throttle button)
43. Open the Hauptschalter, lower the pantograph, Richtungsschalter to 0, reverser key out. (Speedbrake lever) (PARK BRK)
44. Isolate SiFa, close windows and doors, lights off in both cabs and the machine room, battery off. (Left aux)

If it will not move

- Reverser stuck: keep PARK BRK up (the key is held only while it is) and the Fahrschalter at Off. The Fahrschalter moves only at V.
- 'Kein Führerstand besetzt' stays: key in and a direction set. The profile drives the front (F) cab only.
- SiFa or PZB will not come on: open their cut-off valves in the machine room; the aux squares only set the isolation switches.
- AFB brakes freight at 0.3 m/s²; do not let it brake and release in quick succession, the distributors can stick.
- PZB isolated: 100 km/h enforced (50 km/h allowed from 2012). Isolate LZB light or at Brh 65 % or less.

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